RAG Spoofing

Attack, Detection and Defense in Retrieval-Augmented Generation

How an adversary silently corrupts RAG answers — and how to stop it

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Motivating Use Case

Motivation: RAG pipelines retrieve documents from an external corpus and pass them to an LLM — but that corpus can be poisoned

Background: Retrieval-Augmented Generation combines document retrieval with LLM-based answer generation for question answering systems, chatbots, and enterprise knowledge bases.

Use Case: User query → retrieve documents → LLM generates answer.

Problem: An adversary injects synthetic chunks that score high in retrieval but contain no real information — flooding the top-k results and misleading the LLM

<u>Example Attack</u>

User question: What is the capital of Canada?

Top-3 retrieved documents

1. ✗ **Spoofed document (Rank 1)** "Canada is a large country in North America. Toronto is its largest city and a major economic center." .1
2. ✓ **Real document (Rank 2)** "Ottawa is the capital city of Canada." .2
3. ✓ **Real document (Rank 3)** "The capital of Canada is Ottawa, located in Ontario." .3

LLM output: Toronto

Even when the correct answer exists in top-k, spoof documents can dominate the ranking and mislead the LLM.

Formal Problem Statement

Novelty: Spoof chunks must satisfy 3 constraints simultaneously: diverse content, no real information, and embedding closer to the query than real documents

Task Definition

Given a query and a retrieved chunk, determine whether the chunk is a real document or a spoof — and re-rank accordingly

Input

Query + top-k chunks from FAISS/BM25 retrieval (before LLM generation, may include spoof documents).
We do not modify the LLM — we intervene at the retrieval stage.

Output

Re-ranked results with spoofs down-weighted; metrics: Recall drop, Top-1 spoof win rate, Attraction margin

Goal

Quantify retrieval degradation under attack and show that a text-only defense can partially recover it

The challenge: spoof chunks may be more similar (in embedding space) to the query than real documents.

Mission: Text-only defense — 6 suspicion signals, no ground-truth labels required

Models & Methodology

Retrieval Models

Dense: MiniLM, BGE — semantic similarity (FAISS)

Sparse: BM25 — keyword matching

Hybrid: α-weighted fusion for robustness

Pipeline

1. Prepare corpus for retrieval (SQuAD, chunking 120w with 30 overlap, FAISS indexing).
2. Baseline retrieval (dense, BM25, hybrid).
3. Generate spoof documents (5 attack styles).
4. Inject spoof documents into the corpus and rebuild index
5. Retrieval under attack.
6. Apply defense layer (text-based signals, re-ranking).
7. Evaluate and compare results (Recall@k, spoof win rate, LLM judge).

Key Insight

Dense retrieval is more vulnerable to semantic spoofing. BM25 is more robust. Hybrid provides the best baseline but defense must work without labels.

Dataset Construction & Evaluation

01

Dataset Preparation

SQuAD v1.1: 5,000 train + 1,000 validation docs. 120-word chunks with 30-word overlap. FAISS + BM25 indexes built for each condition.

02

Spoof Generation

5 writing styles (encyclopedic, news, textbook, forum, abstract). GPT-4o-mini with temperature 0.9 ensures diversity. Each chunk: relevant keywords but zero real facts.

03

Injection Strategy

300 queries × 5 styles = 1,500 spoof chunks merged into corpus. New index built over augmented corpus for attack evaluation.

04

Evaluation Setup

3 conditions: baseline / under attack / after defense. Metrics: Recall@20, Top-1 spoof win rate, rank displacement, LLM answer correctness (GPT-4o-mini judge).

Note: Recall@20 is used to capture whether the correct answer remains in the candidate set, even if not ranked first.

Metrics and Evaluation Plan

Metrics

Recall@20 — relevant doc in top-20 results
Top-1 spoof win rate — fraction of queries where top-1 is a spoof
Attraction margin — difference between top spoof score and top real document score
Rank displacement — how far real doc falls after injection
LLM judge — GPT-4o-mini: correct / misleading / unsupported

Defense signals

Query overlap ratio
Lexical diversity (TTR)
Surface pattern hits
All signals are derived from query–chunk text only
No ground-truth labels required
but does not fully eliminate spoof influence
False claim density

Expected results

Dense retrieval: Recall drops ~30-50% under attack
BM25: more robust to semantic spoofing
Defense partially recovers Recall without using labels
LLM judge shows higher misleading rate under attack

Key Insight: RAG spoofing is a realistic threat: a text-only defense without ground-truth labels can detect template spoofs. LLM-generated spoofs remain the open challenge.